



inClass- L1

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

Dr. Prakash S Badal

Flow

- Motivation
- Course Outline
- Announcement
- Introduction

What is this course
about?

Physics vs Engineering

I have a steel ball.

I drop it from the second floor.

Ht $\sim 7.2\text{m}$

$t_{\text{descent}} = 1.2\text{s}$

($g = 10\text{ m/s}^2$)

$\Rightarrow$ Suppose we repeat this 100 times
WHY DO WE NOT OBSERVE 1.2s every single time
 $\rightarrow$ human error (release, time measurement)
air-resistance

Physics vs Engineering

Real World

- True mass
- True gravity
- True t_{descent}

Observation

- sensor noise
- precision error
- calibration error

Models

≡ Equations

- Air resistance
- gravity diff.

All Models are Wrong; But some are useful.
George Cox

Prediction

Reality → observations → Model
→ Prediction

Decision Making

PHYSICS: Truth

ENGINEERING: Decision
under uncertainty

The Three Layers

Layer	Question	Discipline
Reality	What is true?	Physics
Observation	What do we know/ see?	Probability & Statistics
Action	What should we do?	Engineer

The Three Layers

Domain	What we have	What we don't know	What we must decide
Engineering	Physics laws Material properties	Loads, earthquakes, deformation	Design a bridge; Inspect a dam.
Justice system	Evidence, forensics	What actually happened, witness reliability	Acquit, punish, investigate more
Medicine			
Disaster management			
Stock markets			

Motivation

Question 1

Think of a
random number
between 1 & 10.

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Karalar et al., 2022. Frontiers

Let's clear the poll. Trivia warm-up!

→ How many players per side in soccer?

11

→ How many stripes on the US flag?

13

→ Atomic Number of P?

15

Question 2

Think of a
random number
between 11 &
20.

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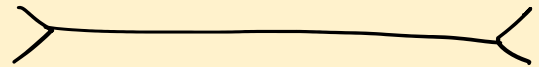
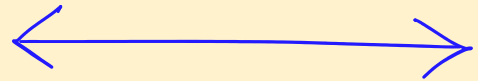
Karalar et al., 2022. Frontiers

What happened?
“attempted PRIMING”

HYPOTHESIS

Thinking, Fast and
Slow

Daniel Kahneman



0. Course Overview

This course is about...

- Making decisions under uncertainty
- Mathematical foundations of probability & statistics
- Cultivating a probabilistic way of thinking
- How to reason with
 - Incomplete information
 - quantify uncertainty
 - Update beliefs with evidence
 - Make rational engineering decisions

Learning outcomes for this course

- Model uncertainty
- Analyze and interpret data
- Make engineering decisions under uncertainty
- Critically evaluate evidence and models
- Develop a probabilistic mindset

Three Steps of Every Engineering Problem

References

Primary References

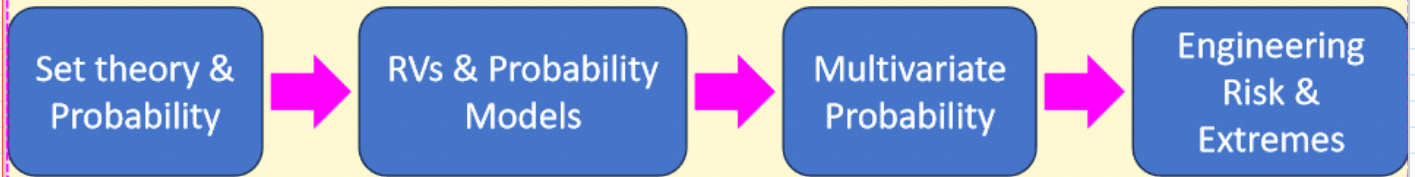
- Benjamin J.R. and Cornell, C.A., Probability, Statistics and Decisions for Civil Engineers, John Wiley, 1970.
- Ang A.H.S. and Tang, W.H., Probability Concepts in Engineering and Design, Vols. 1 and 2, John Wiley, 1975.
- Papoulis, A., Probability, Random Variables and Stochastic Processes, 3rd Ed., McGraw-Hill, 1991.

Recommended Reading

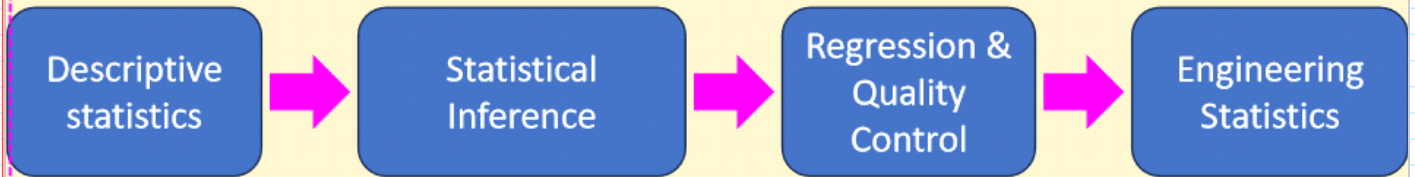
- Kahneman, D. Thinking, Fast and Slow (selected chapters), Penguin, 2011.
- The Black Swan, Nassim Nicholas Taleb, Random House, Penguin, 2013.
- The Signal and the Noise, Nate Silver, 2012

Course Flow

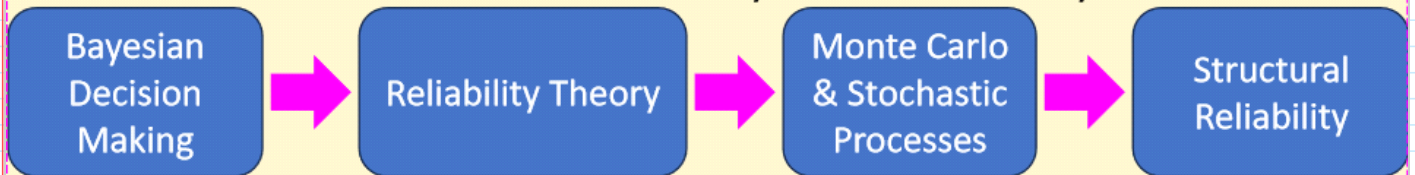
Module 1: Probability – Foundation of Uncertainty



Module 2: Statistics – Learning from Data



Module 3: Decision Analysis and Reliability



Administrative stuff

- Credit: 12
- Lecture hours (Slot A):
 - Venue: BSB 312
 - M 8–8:50 am; T 1–1:50 pm; R 11–11:50 am; F 10–10:50 am
- Tutorial hours
 - Two consecutive hours required
 - **Class rep to confirm**
- Office hours:
 - By appointment via email (STR 305/online)
- Teaching assistants:
 - Pramit Majumder id25d016@smail.iitm.ac.in
 - Shaik Malik Basha ce25m075@smail.iitm.ac.in
 - Swastik Mishra ce25m079@smail.iitm.ac.in

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Assessment



20% + 20%: Quizzes (31st Aug, 12th Oct)



20%: 10-12 Tutorials (No phones, no tabs)



40%: End-semester exam (10th Nov)

Assessment

- Check Moodle regularly (courses.iitm.ac.in)
 - for announcements, book recommendations, YouTube channels, lecture notes, references, and general reads
- ~10-12 tutorials (1.5-2 hours long).
- Final exam
 - Closed book, closed notes

Thank you!